

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

THERMODYNAMICS

LECTURE-03 **PHYSICS**

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Topics To Be Covered

- 1 Revision
- 2 1st Law of thermodynamic
- 3 Work Done By Isothermal Process / A
- 4 ..
- 5 ..

Recap - ①

• Thermodynamics

- System
- Surrounding
- universe

• Intrensic property

• Extrensic property

• Isothermal

• Isobaric

• Isochoric

• Adiabatic

• Reversible

• Irreversible

• Cyclic process

• Temperature

• Thermal equilibrium

• Zeroth law of thermodynamics

Recap from 2nd Lecture

• Heat $Q_p = nC_p \Delta T$ $Q+$ $Q-$

• Internal Energy $U = nC_v \Delta T$ $U+$ $U-$

• Work Expansion $W = +ive$ Compression $W = -ive$

isobaric

• $W = P \Delta V$

Ideal gas equation

$$PV = nRT$$

$$P \Delta V = nR \Delta T$$



lecture no. 3

1st law of thermodynamics

2026, 2025, 2024, 2023 3mark

①

• This law is based on law of Conservation of Energy

• Heat provided to the system
does two things (1st increase) $\uparrow\uparrow$
the internal energy) and does
work done

$$Q_p = nC_p \Delta T$$

$$U = Q_v = nC_v \Delta T$$

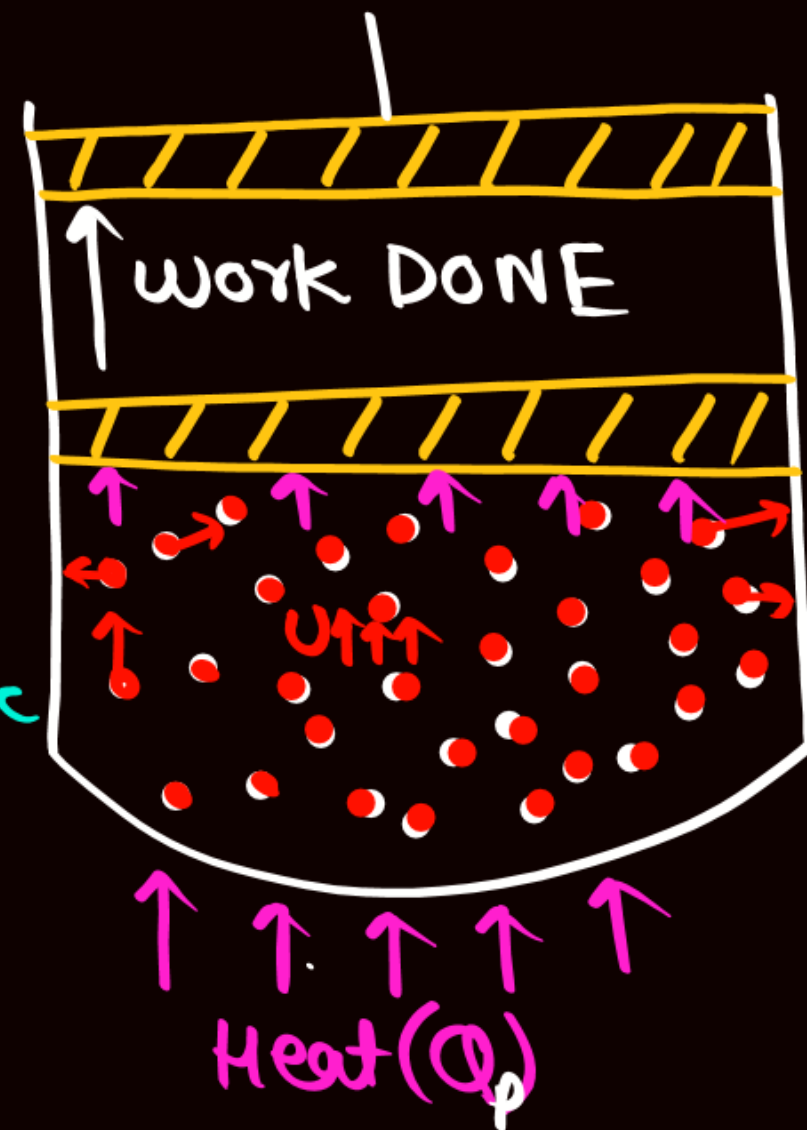
$$W = P \Delta V$$

$$Q_p = U + W$$

$$100J = 75J + 25J$$

$$nC_p \Delta T = nC_v \Delta T + P \Delta V$$

→ Mathematical
Representation
of 1st law
of thermodynamic



Numerical 2025

②

Q $Q = 100 \text{ J}$ $U = 70 \text{ J}$ $W = ?$

(a) 25 J

(b) 30 J

(c) 35 J

(d) 40 J

Relationship between C_p and C_v

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3



⇒ As we know that
from 1st law of
thermodynamic

$$Q_p = U + W$$

$$nC_p \Delta T = nC_v \Delta T + P \Delta V$$

$$nC_p \Delta T = nC_v \Delta T + nR \Delta T$$

$$nC_p \Delta T = n \Delta T (C_v + R)$$

$$C_p = C_v + R$$

$$C_p - C_v = R$$

$$Q_p = nC_p \Delta T$$

$$U = Q_v = nC_v \Delta T$$

$$W = P \Delta V = nR \Delta T$$

$$C_p - C_v = R$$



Heat Capacity Ratio (γ) ^{→ gamma}

2021

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2 mark

4



⇒ It is the ratio of Specific heat capacity at constant pressure to Specific heat capacity at constant Volume

$$\gamma = \frac{C_p}{C_v}$$

→ work done during Adiabatic process.



Note $\Rightarrow C_p - C_v = R$

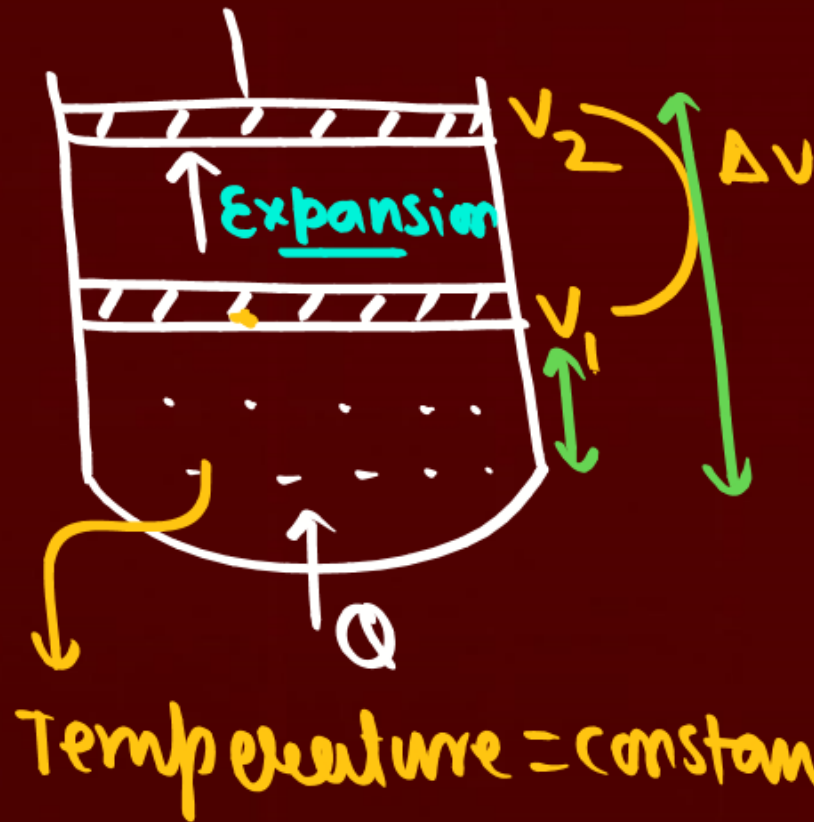
⑤

$$\Rightarrow \frac{C_p}{C_v} = \gamma$$

I

Work Done By Isothermal Process

- During Isothermal Process
Temperature = Constant



$$\# \underline{W} = P \underline{\Delta V}$$

$$d\underline{w} = \underline{P} d\underline{v}$$

$$\underline{d}w = \frac{nRT}{v} \underline{d}v$$

$$PV = nRT$$

$$P = \frac{nRT}{v}$$

on integrating b/s

$$\int d\underline{w} = \int_{v_1}^{v_2} \frac{nRT}{v} dv$$

$$\int d\underline{w} = nRT \int_{v_1}^{v_2} \frac{1}{v} dv$$

$$\int \frac{1}{x} dx = \ln x$$

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Ideal gas equation



$$\int dw = \int_{v_1}^{v_2} \frac{nRT}{v} dv$$

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$$\underline{\int dw = nRT \int_{v_1}^{v_2} \frac{1}{v} dv} \quad \int \frac{1}{x} dx = \ln x$$

$$\text{Work} = nRT [\ln v]_{v_1}^{v_2}$$

$$\text{Work} = nRT [\ln v_2 - \ln v_1] \quad \ln a - \ln b = \ln\left(\frac{a}{b}\right)$$

$$\text{Work} = +nRT \ln\left[\frac{v_2}{v_1}\right]$$

$$W = nRT \ln \left[\frac{V_2}{V_1} \right]$$

$$\underline{\ln} = \underline{2.303} \log$$

⑧

$$W = nRT \underline{2.303} \log \left[\frac{V_2}{V_1} \right]$$

$$W = 2.303 nRT \log \left[\frac{V_2}{V_1} \right]$$

Expansion

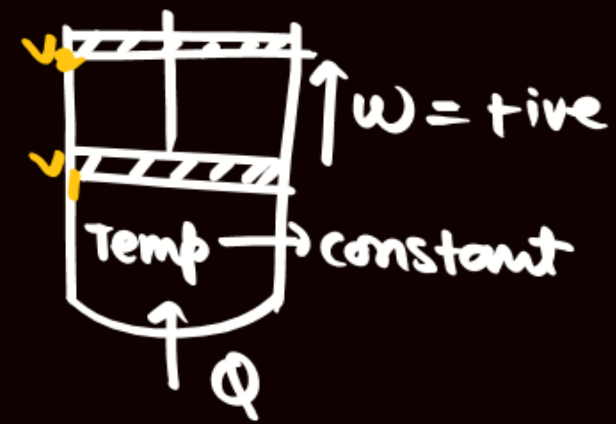
$$W = -2.303 nRT \log \left[\frac{V_2}{V_1} \right]$$

Compression

Repeat

WORK → Temperature = constant

$$W = P \Delta V$$



$$dw = P dv$$

$$PV = nRT$$

$$P = \frac{nRT}{V}$$

$$dw = \frac{nRT}{V} dv$$

$$\int dw = \int \frac{nRT}{V} dv$$

$$W = \int_{V_1}^{V_2} \frac{nRT}{V} dv$$

$$\int \frac{1}{x} dx = \ln x$$

$$W = nRT \int_{V_1}^{V_2} \frac{1}{V} dv$$

$$W = nRT \left[\ln V \right]_{V_1}^{V_2}$$

$$\ln a - \ln b = \ln \frac{a}{b}$$

$$W = nRT \left[\ln V_2 - \ln V_1 \right]$$

$$W = nRT \ln \frac{V_2}{V_1} \quad \ln = 2.303 \log$$

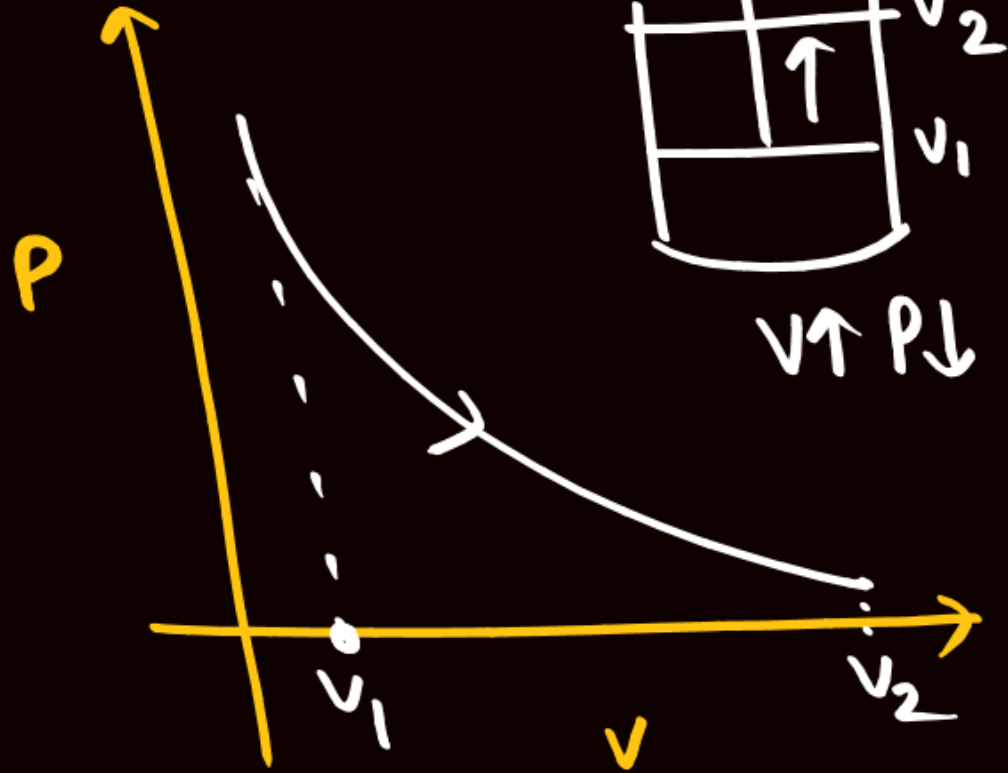
$$W = nRT \cdot 2.303 \log \frac{V_2}{V_1}$$

$$W = +2.303 nRT \log \left(\frac{V_2}{V_1} \right) \text{ Expansion}$$

$$W = -2.303 nRT \log \left(\frac{V_2}{V_1} \right) \text{ Compression}$$

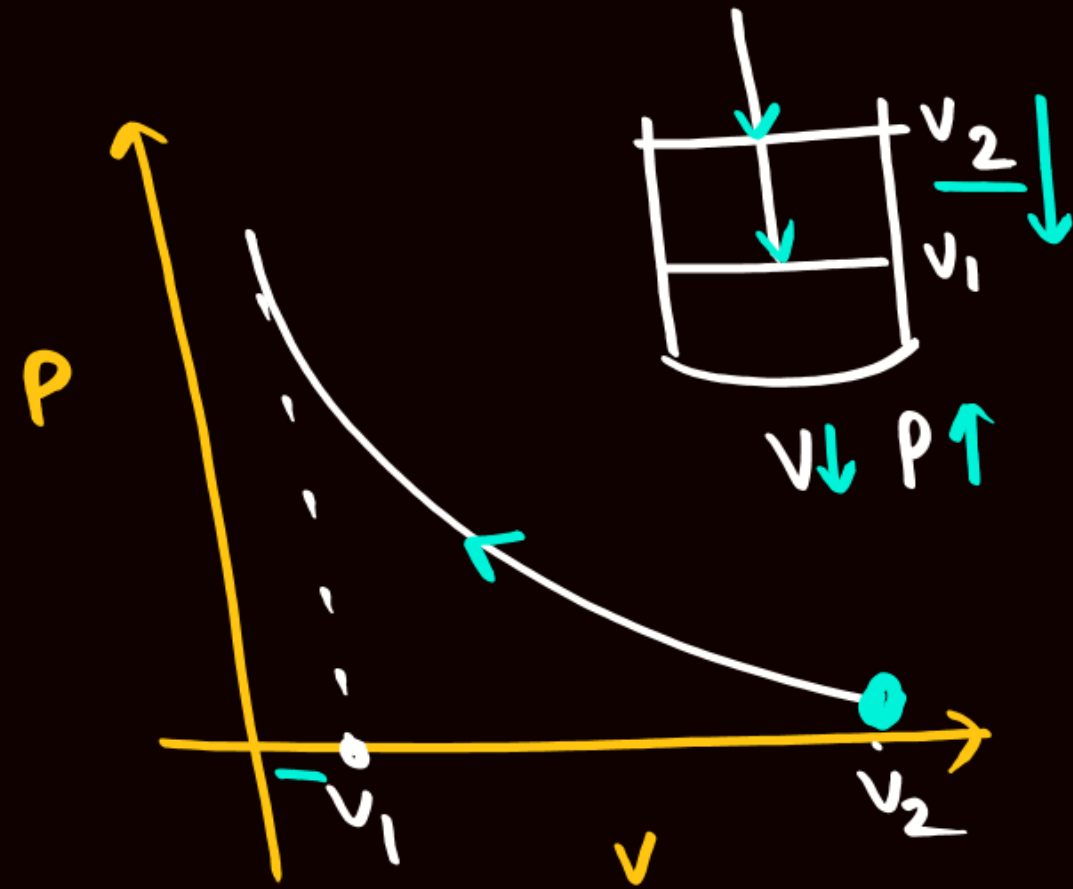
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PV-Graph
Isothermal
 ↓
Neet
2023



Expansion

$$W = +2.303 RT \ln \log \left[\frac{V_2}{V_1} \right] \checkmark$$



Compression

$$W = -2.303 RT \ln \log \left[\frac{V_2}{V_1} \right] \checkmark$$

SUMMARY



$\Rightarrow$ Ist law of thermodynamic
 $Q = U + W$

$$\Rightarrow C_p - C_v = R$$

$$\Rightarrow \gamma = \frac{C_p}{C_v}$$

$$\Rightarrow W = \underbrace{2.303}_{\text{conversion factor}} RT \ln \log \left(\frac{V_2}{V_1} \right)$$

Q1/50



Q. which one is greater

$$\underline{\underline{Q}} \quad \underline{\underline{U}}$$

Q. which one is greater

$$C_p \quad C_v$$

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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